Discussion week 10

University of California, Riverside
1202, 2021 Palindrome Day!

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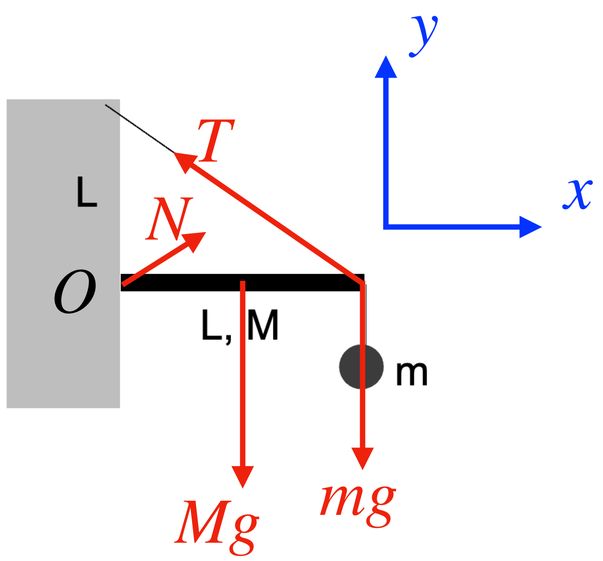
I.

1. A rod (mass M, length L) is sticking out from a building, and it is attached by a hinge. A rock (mass m) is hanging from the end of the rod. To keep the rod from rotating around the hinge (and the rock from falling) a long cable has been attached to the building at a distance of L above the hinge and tied to the end of the rod.



a. What is the tension in the cable? Hint, the rod's mass can all be assumed to be at the center of mass of the rod.

$$\boxed{\sum \tau = 0} \rightarrow Mg \cdot \frac{L}{2} + mg \cdot L - T \cdot \frac{L}{\sqrt{2}} = 0 \rightarrow T = \frac{\sqrt{2}}{2} Mg + \sqrt{2} mg$$

b. What is the normal force of the building on the end of the rod?

$$\boxed{\sum F = 0} \rightarrow \begin{array}{l} x : N_x = T_x = T \cos 45^\circ = 1/2 Mg + mg \\ y : N_y = Mg + mg - T_y = 1/2 Mg \end{array}$$

One can also choose other pivots to find N

2.

2.A race is set up with four different round objects rolling down a ramp (height h). The four objects all have the same mass (M) and radius (R). The shapes are: (1) A solid sphere, (2) a solid cylinder, (3) a cylindrical hoop, and (4) a massless ring with four masses (each with a mass of M/4) spaced evenly around the hoop. Assume no rolling friction. → Energy is conserved

a. Look up what the moments of inertia are for the first three shapes. What is the moment of inertia for the shape #4? $k_1 = 2/5,\ k_2 = 1/2,\ k_3 = 1, k_4 = 1\ (I = kMR^2)$

b. What is the mechanical energy of the objects at the top of the ramp?

$$E = Mgh$$

c. What is the mechanical energy of the objects at the bottom of the ramp?

$$E = Mgh$$

d. The mech. energy at the bottom is kinetic energy. What types of kinetic energy?

Translational kinetic energy and rotational kinetic energy

2.

e. Write an equation for the first object that sets the initial mechanical energy equal to the final kinetic energy.

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

f. Think carefully, how is the angular velocity of the objects related to their velocities?

$$v = \omega R$$

g. Use part f to substitute for angular velocity in the conservation of energy equation.

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I(v/R)^2$$

h. Solve for the final velocity of each object.

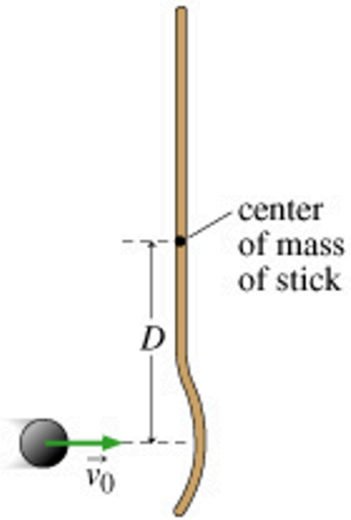
$$v = \sqrt{\frac{2gh}{1 + \frac{I}{MR^2}}} = \sqrt{\frac{2gh}{1+k}}$$

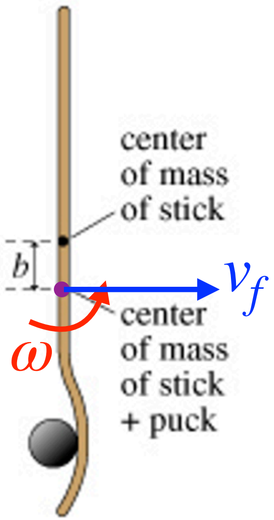
i. Which object reaches the bottom of the ramp with the highest velocity? The lowest?

The one with the smallest k parameter.

3.

3.A hockey stick of mass m_s and length L is at rest on the ice (which is assumed to be frictionless). A puck with mass m_p hits the stick a distance D from the middle of the stick. Before the collision, the puck was moving with speed v_0 in a direction perpendicular to the stick, as indicated in the figure. The collision is completely inelastic, and the puck remains attached to the stick after the collision.





Translational motion+ rotational motion

3.

a. Find the speed v_f of the center of mass of the stick+puck combination after the collision.

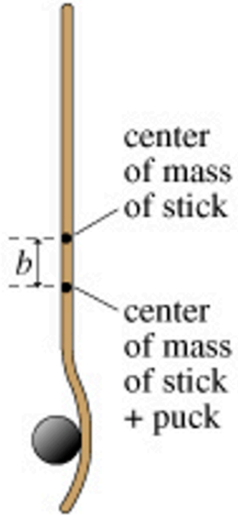
Conservation of momentum

$$m_p v_0 = (m_p + m_s) v_f \rightarrow v_f = \frac{m_p v_0}{m_p + m_s}$$

b. After the collision, the stick and puck will rotate about their combined center of mass. How far is this center of mass from the point at which the puck struck? In the figure, this distance is (D−b).

$$\vec{r}_{cm} = \sum_i \frac{m_i \vec{r}_i}{m_{tot}}$$

$$m_p D = (m_s + m_p) b \rightarrow b = \frac{m_p D}{m_s + m_p} \qquad D - b = \frac{m_s D}{m_s + m_p}$$



c. What is the angular momentum L_{cm} of the system before the collision, with respect to the center of mass of the final system?

$$L = m_p v_0 (D - b) = \frac{m_s m_p}{m_s + m_p} v_0 D$$

d. What is the angular velocity ω of the stick+puck combination after the collision? Assume that the stick is uniform and has a moment of inertia I_0 about its center.

$$\omega = L / I_0$$